



Subchronic oral toxicity study of Korean red ginseng extract in Sprague-Dawley rats with a 4-week recovery period

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ABSTRACT

Ginseng is a major herbal remedy used in Asian countries for thousands of years and known to restore and enhance vital energy. Korean red ginseng, which is processed by steaming and drying fresh *Panax ginseng*, is most popular and contains unique ginsenosides, which have anticancer and anti-inflammatory properties. The present study was carried out to evaluate the repeated oral dose toxicity of Korean red ginseng extract. The test article was administered orally once a day to male and female Sprague-Dawley rats at dose levels of 0, 500, 1000, or 2000 mg/kg/day for 13 consecutive weeks (15 animals/sex/group in the vehicle control and 2000 mg/kg/day groups, and 10 animals/sex/group in the 500 and 1000 mg/kg/day groups). Ten animals per group were sacrificed at the end of the 13-week treatment period, and the remaining rats were sacrificed after a 4-week recovery period. Administration of Korean red ginseng extract did not result in any toxicologically significant changes in mortality, body weight, food consumption, ophthalmoscopy, hematology, serum biochemistry, gross pathological findings, absolute/relative organ weights, or histopathology. It was established that the no observed adverse effect level (NOAEL) of the test article was 2000 mg/kg/day for both sexes in this study.

1. Introduction

Ginseng is a major herbal remedy in traditional medicine of eastern Asia, and has been used for over two thousand years in many countries to restore and enhance vital energy (Chong and Oberholzer, 1988). Ginseng-based supplements are common and popular herbal supplements; a national survey of men and women in the USA reported that 3% of the population used ginseng preparations in 2002 (Kaufman et al., 2002).

It is well documented that the efficacy of ginseng is mostly attributed to ginsenosides, also known as ginseng saponin. These major components are responsible for the biochemical and pharmacological effects of ginseng (Attele et al., 1999; Liao et al., 2002; Shi et al., 2010; Yue et al., 2006). To date, more than 150 naturally occurring ginsenosides have been identified from *Panax* species, and individual ginsenosides may have different properties (Shibata, 2001). In general, ginsenosides are categorized into 20 (S)-protopanaxadiol and 20 (S)-protopanaxatriol groups. Rb1 and Rg1 are major ingredients of ginseng, and belong to the 20 (S)-protopanaxadiol and 20 (S)-protopanaxatriol groups, respectively (Shin et al., 2006). Rb1 and Rg1 rescue spinal cord neurons from death induced by excitotoxicity and oxidative stress, and

reduce apoptotic neuron death (Mochizuki et al., 1995). Furthermore, Rb1 and Rg1-injected mice exhibit enhanced spatial learning abilities by increasing hippocampal synaptic densities without changing the plasticity of individual synapses (Nam, 2002).

There are several species of ginseng, including *Panax ginseng* (Korea), *Panax notoginseng* (China), *Panax japonicas* (Japan), and *Panax quinquefolius* (America). *Panax ginseng* is the most valued species, and has been reported to contain larger amounts of ginsenosides than others (Choi, 2008). In Korea, there are several traditional ways of preparing and manufacturing *Panax ginseng*. For medical purposes, ginseng is harvested when it is six years old. Red ginseng is steamed and then dried without peeling by a traditional method, whereas white ginseng is simply prepared by drying and peeling the roots. Unlike white ginseng, the processed red ginseng contains ginsenosides Rg3, Rg5, and Rk1, and these distinct ginsenosides have potent biological properties, such as anticancer and anti-inflammatory activities (Lee et al., 1999; Mook-Jung et al., 2001; Xie et al., 2006). In addition, Korean red ginseng has been shown to exert positive pharmacological effects on high blood pressure, atherosclerosis, and hyperlipidemia by reducing oxidative damage (Kim and Rhee, 2009; Nocerino et al., 2000). These positive effects may be explained by increased levels of nitric oxide (NO), which

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acts as a natural vasodilator. Ginsenosides of Korean red ginseng enhance the formation of citrulline, which is a precursor of arginine, and arginine is needed to produce NO (Chen and Gillis, 1992; Chen et al., 1984; Gillis, 1997; Kim et al., 1994). Furthermore, male volunteers administered Korean red ginseng extract showed increased NO levels in exhaled breath, along with decreased blood pressures and heart rates (Han et al., 2005).

The US Food and Drug Administration (FDA) reported adverse effects from products containing ginseng, and signs and symptoms of overexposure to ginseng preparations are termed ginseng abuse syndrome. This condition is characterized by hypertension, together with nervousness, sleeplessness, skin eruptions, edema, and diarrhea (Qi et al., 2010; Chan and Fu, 2007). However, when toxicities due to ginseng preparations were reported, sample preparations varied among different manufacturers. In addition, there were differences among batches in terms of content, quality, and preparation method used for each preparation, indicating the difficulty of evaluating ginseng preparation toxicity in humans (Chan and Fu, 2007).

Despite the long history and wide consumption of ginseng, the chronic effects are not well characterized, and little information on toxicity is available on any specific ginseng preparation (Chan and Fu, 2007). To obtain a more comprehensive understanding of ginseng preparation toxicities, we assessed the toxicity of Korean red ginseng extract on Sprague-Dawley (SD) rats.

2. Materials and methods

2.1. Experimental design

2.1.1. Test article

Korean red ginseng extract, the test material, was provided by the Korea Ginseng Corp. (Daejeon, Korea). The extract used for this experiment was prepared as described in Lee et al. and used the same batch with a previous experiment, as described in Park et al. (Park et al., 2013; Lee et al., 2015). The ginsenoside components present in Korean red ginseng extract was measured by UPLC and reported in Park et al. which previous reported experiment using same batch of test material (Park et al., 2013). The raw botanical material was analyzed by Dionex ICS-5000 Bio-LC system (Thermo Scientific, NY, USA). The appropriate quantity of extract was weighed and suspended in sterile distilled water for the high dose preparation. The high dose suspension was then serially diluted in sterile distilled water (Daehan Pharm, Korea) to prepare it for administering to the middle and low dose groups. The dosing suspensions were prepared at least once every day during the treatment. The test article and vehicle control were forcibly administered by oral gavage with a syringe for 13 weeks. Individual dosing volumes (10 mL/kg body weight) were calculated based on the most recently recorded body weight, using the Path/Tox System (version 4.2.2, Xybion Medical Systems Corporation, USA).

2.1.2. Experimental animals

Male and female SD rats (CrI:CD, 5 weeks old) were obtained from Orient Bio Ltd. (Gyeonggi-Do, Korea) and acclimated for 1 week prior to commencement of the experiment. The animals were housed at a temperature of 23 ± 3 °C, relative humidity of $50 \pm 10\%$, air ventilation of 10–20 times/hour, photoperiod of 12-h/day (08:00 light on ~20:00 light off), and light intensity of 150–300 Lux. Throughout the study, the temperature and relative humidity of the animal room were controlled and recorded automatically. Environmental conditions such as ventilation and light intensity were periodically monitored, showing that the room temperature was generally in the range 20–26 °C and relative humidity was generally in the range 40–60%. Animals

(four animals for the quarantine and acclimation periods, one and two animals per cage for the pretreatment, treatment and recovery periods) were maintained in a stainless steel wire cage (255 W × 465 L × 200 H mm). Pelleted food (Labdiet #5053, PMI Nutrition International, USA) and water were sterile and available *ad libitum*. No significant deviations that would be expected to affect the outcome of the study were observed. The present study was performed in compliance with the Good Laboratory Practice (GLP) of the Organization for Economic Cooperation and Development and the Korea Food and Drug Administration. The protocols for animal studies were reviewed by the Institutional Animal Care and Use Committee (IACUC) of the Korea Institute of Toxicology (KIT).

2.1.3. Inoculation

As a preliminary 4-week study of Korean red ginseng extract in SD rats at dose levels of 500, 1000, and 2000 mg/kg/day, there were no treatment-related toxicity findings (Park et al., 2013). From these results, doses of Korean red ginseng extract were determined to be 500, 1000, and 2000 mg/kg/day for both males and females for this study. Sterile distilled water was administered to the vehicle control group. The animals were randomly divided into 4 groups per sex, consisting of 10/sex per each group for the 13-week treatment period, and 5/sex in the vehicle and 2000 mg/kg/day groups for the recovery period.

2.2. Animal observation

2.2.1. Clinical signs and mortality

All animals were observed once a day during acclimation, pretreatment, and recovery periods and twice a day during the treatment period for clinical findings and mortality. The clinical signs were recorded using the Path/Tox System (version 4.2.2). If abnormal signs were observed, the type and frequency of symptoms and the day of symptom expression were recorded. Detailed observation for the severity of symptom was recorded once during pretreatment period and once a week during the treatment and recovery periods.

2.2.2. Body weight and food consumption

The body weight of each animal was measured at receipt, grouping, and then approximately once a week during the treatment period and in the last week of treatment and on the scheduled day of necropsy. Food consumption was measured once before treatment and approximately once a week during the treatment and recovery periods. Measured amounts of food were supplied to each cage, and the remaining food was measured on the next day to calculate the difference, which was taken to be daily food consumption (g/rat/day).

2.2.3. Ophthalmological examinations

Ophthalmological examination was conducted in all animals during the pretreatment period after grouping, in all main and recovery animals during the treatment period, and in all recovery animals during the last week of the recovery period. Both eyes were examined externally and then by a binocular indirect ophthalmoscope (Vantage Plus Digital, Keeler Ltd., England) after a mydriatic treatment (Mydrin-P, Santen Pharmaceutical Co., Japan).

2.3. Laboratory investigations

2.3.1. Hematology

All animals were fasted overnight prior to necropsy and blood collection. Approximately 1.5 mL of blood was drawn from the posterior vena cava of all animals terminated at scheduled necropsy days under isoflurane anesthesia. The following items were then measured by a

Table 1

Mortality of male and female SD rats orally administrated with Korean red ginseng extract for 13-week and 4-week recovery studies.

Dose (mg/kg/day)	Dosing period (13-week)		Recovery period (4-week)	
	Male	Female	Male	Female
0	0/15	0/15	0/5	0/5
500	0/10	0/10		
1000	0/10	0/10		
2000	0/15	0/15	0/5	0/5

Table 2

Clinical signs of male and female SD rats orally administrated with Korean red ginseng extract for 13-week and 4-week recovery studies.

Sex	Male						Female					
Dose (mg/kg/day)	Control	500	1000	2000	Control (R)	2000 (R)	Control	500	1000	2000	Control (R)	2000 (R)
No. of animals	15	10	10	15	5	5	15	10	10	15	5	5
Loss of fur	2	2	1	4	1	2	1	0	1	0	1	0
Scratch wound	1	0	0	3	1	0	0	0	0	0	0	0
Hard skin	0	0	0	0	0	0	0	0	0	1	0	0
Scab	0	0	0	2	0	1	0	0	0	0	0	0
Loss of teeth	1	0	0	0	0	0	0	0	0	0	0	0
Eye discharge	0	0	0	0	0	0	1	0	0	1	0	0

(R): Recovery period.

hematological auto-analyzer (ADVIA120 Hematology System, Bayer, USA): white blood cell count (WBC), red blood cell count (RBC), hemoglobin concentration (HGB), hematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), platelets (PLT), neutrophils (NEU), lymphocytes (LYM), monocytes (MON), eosinophils (EOS), basophils (BAS), large unstained cells (LUC), and reticulocytes (RET). Prothrombin time (PT) and activated partial thromboplastin time (APTT) were measured by a blood coagulation analyzer (ACL 9000 Plus, Instrumentation Laboratory, Italy), after treatment with 3.2% sodium citrate.

2.3.2. Serum biochemistry

Blood was drawn from the posterior vena cava of all animals in the same way as for hematology, and then centrifuged (approximately 3000 rpm, 10 min). The following items were measured using an automated analyzer (200FR NEO, Toshiba Co., Japan): aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), blood urea nitrogen (BUN), creatinine (CREA), glucose (GLU), total cholesterol (TCHO), albumin/globulin ratio (A/G), total protein (TP), albumin (ALB), creatine kinase (CK), triglyceride (TG), total bilirubin (TBIL), phospholipid (PL), calcium (Ca), inorganic phosphorus (IP), chloride (Cl), sodium (Na), and potassium (K).

2.3.3. Urinalysis

Urinalysis was carried out on urine samples collected for approximately 17 h within 1 week prior to the necropsy from all main and recovery animals, respectively. The following items including urine volume, glucose (GLU), bilirubin (BIL), specific gravity (SG), ketone (KET), occult blood (BLO), pH, protein (PRO), urobilinogen (URO), nitrite (NIT) and color were measured. Items except for urine volume

and color were measured by a urine stick (Multistix 10SG, Siemens) and automated urine analyzer (CliniTek-500, Bayer). The parameters were measured by reading the scales of the urine collection tube for urine volume, visual examination for urine color and microscopic examination for urine sediment, respectively.

2.4. Gross findings

After the end of the treatment and recovery periods, all animals had been fasted over night to scheduled necropsy and were terminated by exsanguinations of aorta and posterior vena cava isoflurane anesthesia. Gross post mortem examinations were conducted on terminated animals and organs were collected for weight measurement and histopathological examination.

2.5. Organ weights

Absolute organ weights were measured using an electronic scale, and their relative organ weights (organ-to-body weight ratios) were calculated from the fasted terminal body weight of all animals until sacrifice.

2.6. Histopathological examination

The organs, except for testes, epididymides, and eyes, obtained from all animals were fixed with 10% neutral buffered formalin solution. The testes and epididymides were preserved in Bouin's fixative, and the eyes were fixed in Davidson's fixative for approximately 48 h, and then transferred to 70% ethanol. Paraffin embedding and section were conducted of all fixed tissues of the vehicle control and high dose groups. The sections were stained with Hematoxylin-Eosin (H&E) for histopathological examination.

(A) Fig. 1. Body weight changes of male and female SD rats orally administered with Korean red ginseng extract for 13-week and 4-week recovery studies.

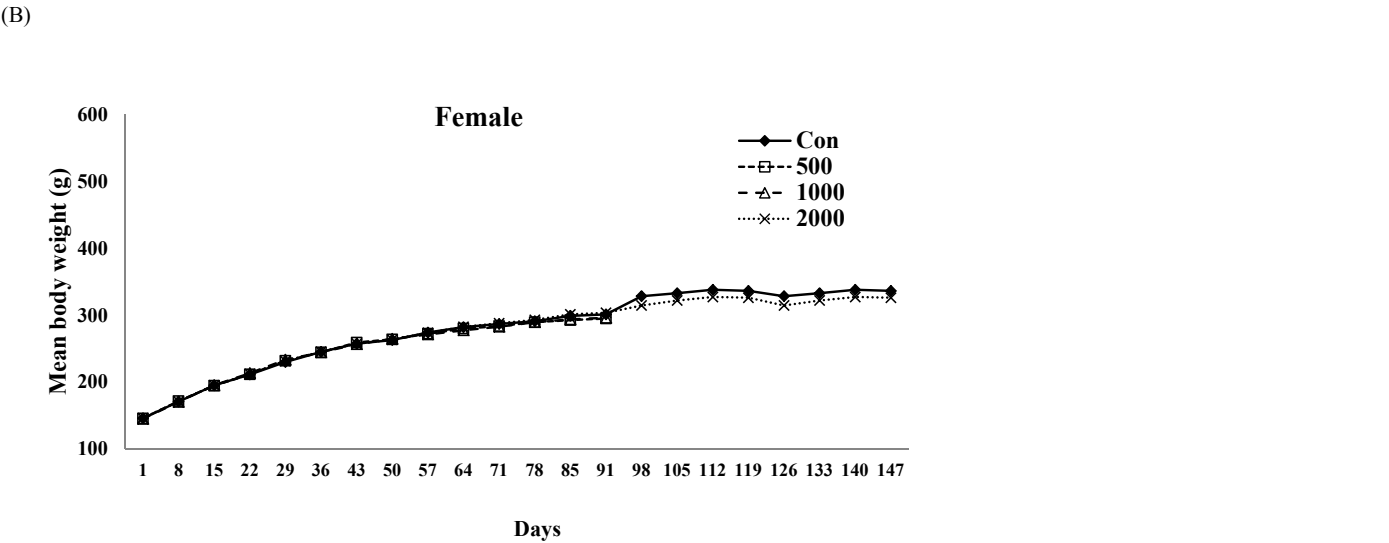
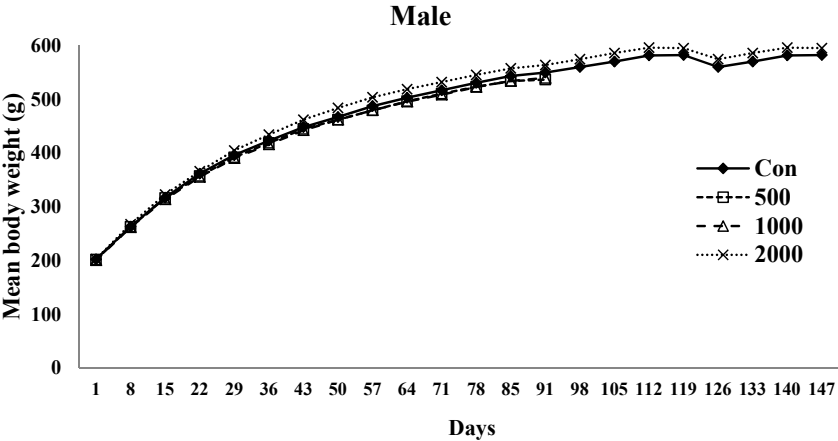


Table 3
Food consumption of male and female SD rats orally administrated with Korean red ginseng extract for 13-week and 4-week recovery studies.

Day	Male				Female			
	Con	500	1000	2000	Con	500	1000	2000
3	28.1 ± 2.37	26.9 ± 1.3	27.2 ± 1.6	27.8 ± 1.86	20.4 ± 1.67	20.6 ± 1.61	20 ± 1.14	20.8 ± 2.18
8	31 ± 2.42	30.8 ± 1.71	29.8 ± 1.56	32.1 ± 2.93	19.7 ± 1.7	20 ± 1.93	19.7 ± 1.29	19.7 ± 1.36
15	32.7 ± 3.43	32.1 ± 3.86	31.4 ± 3.38	31.4 ± 1.6	22 ± 2.49	21 ± 2.09	20.6 ± 1.66	22.1 ± 3.19
22	33.4 ± 2.28	33.1 ± 4.26	30.7 ± 2.44	33.4 ± 3.91	19.1 ± 2.89	19.9 ± 3.04	20.9 ± 2.63	20.8 ± 3.46
29	35.1 ± 2.48	33.5 ± 4.74	32.7 ± 1.49	36 ± 4.17	22.6 ± 4.47	22.5 ± 1.65	23.2 ± 0.55	22.2 ± 2.33
36	31.5 ± 1.65	32.1 ± 4.89	31 ± 2.49	34.6 ± 5.08	22.5 ± 2.47	21.6 ± 1.01	21.6 ± 2.23	22.3 ± 2.91
43	36.4 ± 2.77	36.9 ± 6.92	36.5 ± 3.12	38.5 ± 4.1	26.4 ± 4.83	25.4 ± 2.29	26.4 ± 2.4	27.2 ± 2.12
50	30.5 ± 1.69	32 ± 3.25	31.5 ± 2.59	33.3 ± 4.88	20.5 ± 5.12	20.1 ± 3.97	21.3 ± 2.56	20.4 ± 2.89
57	33.6 ± 2.52	31.5 ± 2.34	30.8 ± 2.05	33 ± 2.93	22.5 ± 4.47	20.5 ± 2.06	21 ± 1.73	21.9 ± 1.25
64	33.4 ± 3.19	32.7 ± 3.03	31.7 ± 2.96	33.8 ± 3.46	22.9 ± 2.92	22.8 ± 1.62	22.8 ± 1.5	23.8 ± 2.66
71	32.5 ± 2.02	32.7 ± 2.84	31.5 ± 3.62	33.9 ± 2.83	23 ± 4.03	21.3 ± 1.31	22 ± 1.78	23.3 ± 2.84
78	31.7 ± 2.06	33 ± 4.19	32 ± 3.18	34.4 ± 3.75	20.9 ± 4.98	22.4 ± 3.31	22.8 ± 1.6	21.6 ± 3.22
85	33.2 ± 4.41	33.6 ± 5.11	33.2 ± 2.1	33.5 ± 3.6	23.3 ± 3.5	22.9 ± 2.47	21.9 ± 1.46	23.6 ± 2.61
91	30 ± 2.19	28.3 ± 4.12	28.1 ± 3.26	29.5 ± 4.57	20.9 ± 2.52	18.8 ± 1.65	20.6 ± 1.77	20.8 ± 2.07
98 (R)	32.4 ± 1.32			34 ± 2.91	25.1 ± 3.67			23.6 ± 1.14
105 (R)	31.1 ± 2.47			34.4 ± 3.29	24.9 ± 5.59			23.1 ± 4.12
112 (R)	32.3 ± 3.81			34.4 ± 4.48	25.2 ± 4.39			22.8 ± 3.89
119 (R)	28.6 ± 1.04			29.6 ± 3.58	20.3 ± 5.4			21 ± 3.41

(R): Recovery period.

Table 4

Hematological values of male and female SD rats orally administrated with Korean red ginseng extract for 13-week and 4-week recovery studies.

Parameter	Dose group (mg/kg/day)					
	Control	500	1000	2000	Con (R)	2000 (R)
Males						
WBC (x10 ⁹ /L)	9.94 ± 1.894	9.66 ± 2.037	10.13 ± 2.303	10.36 ± 2.661	10.77 ± 1.723	12.61 ± 2.223
RBC (x10 ¹² /L)	9.11 ± 0.312	9.21 ± 0.276	9.31 ± 0.36	9.22 ± 0.406	9.44 ± 0.459	9.42 ± 0.103
HGB (g/dl)	16.1 ± 0.39	16.5 ± 0.56	16.3 ± 0.57	16.5 ± 0.75	16.4 ± 0.58	16.3 ± 0.41
HCT (%)	48.7 ± 1.26	49.7 ± 1.51	49.5 ± 2.01	50 ± 2.25	50.9 ± 2.32	50.4 ± 0.99
MCV (fl)	53.5 ± 1.22	54.0 ± 1.17	53.2 ± 1	54.2 ± 1.09	53.9 ± 2.28	53.5 ± 0.88
MCH (pg)	17.7 ± 0.55	18 ± 0.41	17.6 ± 0.27	17.9 ± 0.52	17.3 ± 0.64	17.3 ± 0.41
MCHC (g/dl)	33.0 ± 0.37	33.3 ± 0.43	33.0 ± 0.41	33.0 ± 0.6	32.2 ± 0.53	32.4 ± 0.25
PLT (x10 ⁹ /L)	1187 ± 146.3	1172 ± 180.9	1108 ± 89.7	1104 ± 121.6	1118 ± 82.3	1093 ± 162.3
RET (%)	2.25 ± 0.277	1.98 ± 0.259	2.23 ± 0.336	2.12 ± 0.297	2.26 ± 0.592	2.24 ± 0.222
NEU (%)	15.7 ± 3.75	12.9 ± 3.28	16.9 ± 4.24	12.3 ± 2.95	13.8 ± 3.96	10.3 ± 2.19
LYM (%)	79.6 ± 4.47	82.6 ± 3.69	77.8 ± 4.48	83.1 ± 2.97	81.5 ± 5.08	84.6 ± 2.01
EOS (%)	1.0 ± 0.33	1.1 ± 0.36	1.2 ± 0.35	1.1 ± 0.28	1.1 ± 0.6	1.2 ± 0.23
MON (%)	2.2 ± 0.71	2.1 ± 0.34	2.8 ± 0.66*	2.0 ± 0.35	2.5 ± 0.92	2.3 ± 0.23
BAS (%)	0.4 ± 0.15	0.4 ± 0.18	0.4 ± 0.09	0.4 ± 0.13	0.3 ± 0.08	0.4 ± 0.05
LUC (%)	1.2 ± 0.46	1.0 ± 0.31	1.0 ± 0.56	1.1 ± 0.29	0.7 ± 0.16	1.2 ± 0.54
PT (sec)	15.9 ± 0.61	16.9 ± 1.06**	16.1 ± 0.58	16.3 ± 0.61	14.9 ± 0.33	15.5 ± 0.37*
APTT (sec)	13.8 ± 1.96	15.8 ± 1.55	15.5 ± 1.05	15.0 ± 2.18	13.5 ± 1.12	13.6 ± 0.86
Females						
WBC (x10 ⁹ /L)	6.92 ± 1.64	6.92 ± 1.326	7.94 ± 2.575	8.09 ± 2.809	8.29 ± 1.62	7.72 ± 1.142
RBC (x10 ¹² /L)	8.45 ± 0.268	8.38 ± 0.251	8.24 ± 0.405	8.33 ± 0.258	8.51 ± 0.351	8.5 ± 0.139
HGB (g/dl)	16.0 ± 0.43	15.7 ± 0.43	15.5 ± 0.66	15.7 ± 0.56	16.1 ± 0.58	15.7 ± 0.34
HCT (%)	46.8 ± 1.52	46.2 ± 0.92	45.6 ± 2.11	45.6 ± 1.59	48.3 ± 1.78	47.6 ± 0.69
MCV (fl)	55.4 ± 1.08	55.2 ± 1.31	55.4 ± 0.87	54.7 ± 1.32	56.8 ± 1.07	56.0 ± 0.9
MCH (pg)	18.9 ± 0.33	18.7 ± 0.44	18.9 ± 0.27	18.8 ± 0.43	18.9 ± 0.45	18.4 ± 0.45
MCHC (g/dl)	34.1 ± 0.32	33.9 ± 0.65	34.1 ± 0.49	34.4 ± 0.61	33.2 ± 0.22	32.9 ± 0.48
PLT (x10 ⁹ /L)	1158 ± 136.9	1084 ± 78.1	1123 ± 107.8	1136 ± 154.3	1086 ± 101.7	1109 ± 176.9
RET (%)	2.06 ± 0.268	2.12 ± 0.487	2.00 ± 0.339	2.22 ± 0.586	2.05 ± 0.332	1.80 ± 0.323
NEU (%)	13.1 ± 4.82	11.9 ± 5.03	11.6 ± 5.15	14.6 ± 5.07	10.4 ± 4.40	9.4 ± 2.27
LYM (%)	81.4 ± 5.08	83.2 ± 5.00	83.4 ± 5.29	80.5 ± 5.57	83.9 ± 5.74	85.0 ± 3.41
EOS (%)	1.1 ± 0.25	1.1 ± 0.31	1.2 ± 0.67	1.2 ± 0.33	1.2 ± 0.38	1.2 ± 0.11
MON (%)	2.4 ± 0.68	2.1 ± 0.71	2.1 ± 0.52	2.3 ± 0.80	3.0 ± 0.71	2.5 ± 0.78
BAS (%)	0.4 ± 0.10	0.4 ± 0.25	0.4 ± 0.12	0.4 ± 0.07	0.2 ± 0.10	0.4 ± 0.19
LUC (%)	1.6 ± 0.58	1.4 ± 0.46	1.4 ± 0.53	1.1 ± 0.29	1.3 ± 0.67	1.5 ± 0.82
PT (sec)	16.4 ± 0.64	16.9 ± 0.51	16.3 ± 0.70	16.4 ± 0.83	15.7 ± 0.33	15.7 ± 0.29
APTT (sec)	12.5 ± 0.69	12.4 ± 0.98	12.6 ± 1.28	12.6 ± 1.33	12.4 ± 1.07	12.8 ± 0.92

Values are means ± S.D.

(R): Recovery period.

*: Significant difference compared with control ($p < 0.05$).**: Significant difference compared with control ($p < 0.01$).

2.7. Statistical analysis

For the data obtained from the vehicle control and treated groups, multiple comparison tests were conducted. Homogeneity of variance was analyzed using the Bartlett Test. If it was not significant in the Bartlett Test, the ANOVA test was conducted and, when significant, Dunnett's test was used to determine the difference between the control and treatment groups. If the homogeneity variance was significant in the Bartlett Test, the Kruskal-Wallis (H) Test was conducted and, when significant, the Dunn's Rank Sum Test was conducted to determine the difference between the control and treatment groups. To compare the control group and the recovery group, the data were analyzed using an F-Test for variance homogeneity, and *t*-test was conducted to determine if the recovery group was significantly different from the control group ($\alpha = 0.05$). Dunnett's test was used to determine the difference between the control group and the recovery group. If significant deviations from variance homogeneity were observed, then the Kruskal-Wallis H Test was conducted ($\alpha = 0.05$). If a significant difference was

observed in the Kruskal-Wallis H Test, Dunn's Rank Sum Test was conducted to quantify the specific pairs from group comparison that were significantly different. For the data of frequency and percentage, Fisher's Exact Probability Test was conducted to determine the pairs from group comparison for the significance. Statistical analyses were performed using Path/Tox System (version 4.2.2) at the significance level of $p < 0.05$ or 0.01.

3. Results

3.1. Components of ginsenosides

The red ginseng extract contained 2.18% of total ginsenosides, 1.52% of saccharide, 6.09% of polysaccharide and 3.6% of arginine-fructose-glucose. The detailed ginsenoside contents were described in the previous experiment (Park et al., 2013).

Table 5

Biochemical serum values of male and female SD rats orally administered with Korean red ginseng extract for 13-week and 4-week recovery studies.

Parameter	Dose group (mg/kg/day)					
	Control	500	1000	2000	con (R)	2000 (R)
Males						
GLU (mg/dL)	130.7 ± 22.55	113.1 ± 18.24	124.3 ± 24.63	127.5 ± 24.08	128.2 ± 30.93	136.9 ± 36.62
BUN (mg/dL)	13.6 ± 1.99	14.0 ± 1.06	15.3 ± 1.21	14.0 ± 1.58	15.5 ± 1.09	16.0 ± 1.35
CREA (mg/dL)	0.59 ± 0.035	0.55 ± 0.044	0.57 ± 0.039	0.59 ± 0.034	0.60 ± 0.048	0.62 ± 0.055
TP (g/dL)	6.90 ± 0.285	6.93 ± 0.310	7.08 ± 0.337	7.07 ± 0.269	7.05 ± 0.222	7.17 ± 0.248
ALB (g/dL)	4.29 ± 0.108	4.29 ± 0.122	4.34 ± 0.168	4.34 ± 0.120	4.37 ± 0.066	4.40 ± 0.089
A/G (ratio)	1.65 ± 0.091	1.64 ± 0.124	1.59 ± 0.083	1.60 ± 0.146	1.63 ± 0.102	1.60 ± 0.111
TCHO (mg/dL)	66.9 ± 9.33	60.6 ± 14.58	66.1 ± 16.60	76.7 ± 18.53	68.4 ± 15.08	70.8 ± 18.74
TG (mg/dL)	45.6 ± 14.30	35.1 ± 9.61	55.6 ± 26.26	52.1 ± 25.65	49.1 ± 6.55	51.5 ± 14.57
PL (md/dL)	96 ± 12.0	85 ± 12.4	97 ± 18.2	105 ± 20.2	100 ± 15.5	100 ± 17.2
AST (IU/L)	120.3 ± 19.49	133.9 ± 7.86	118.7 ± 20.83	125.4 ± 14.82	141.2 ± 36.71	160.6 ± 112.64
ALT (IU/L)	34.3 ± 4.65	38.9 ± 5.50	36.1 ± 5.15	36.9 ± 10.87	43.3 ± 14.84	55.1 ± 48.67
TBIL (mg/dL)	0.115 ± 0.0106	0.114 ± 0.0147	0.112 ± 0.0109	0.118 ± 0.0101	0.145 ± 0.0192	0.149 ± 0.0154
ALP (IU/L)	274.1 ± 58.83	239.9 ± 27.19	247.3 ± 36.13	240.9 ± 41.24	210.6 ± 24.85	215.1 ± 14.79
CK (IU/L)	690 ± 191.3	693 ± 92.2	607 ± 205.6	638 ± 184.4	746 ± 288.5	690 ± 362.2
Ca (mg/dL)	11.25 ± 0.436	10.97 ± 0.409	11.18 ± 0.439	11.29 ± 0.244	11.78 ± 0.218	11.92 ± 0.504
IP (mg/dL)	9.22 ± 0.717	8.42 ± 0.847*	8.29 ± 0.557*	8.68 ± 0.646	8.59 ± 0.547	9.13 ± 0.983
Na (mmol/L)	145 ± 1.1	146 ± 1.6	145 ± 1.8	146 ± 1.4	147 ± 1.9	148 ± 1.7
K (mmol/L)	7.85 ± 1.193	7.52 ± 1.068	7.51 ± 1.257	7.97 ± 1.054	7.81 ± 1.312	7.84 ± 1.441
Cl (mmol/L)	102 ± 1.5	102 ± 1.6	101 ± 0.9	102 ± 1.0	103 ± 1.3	103 ± 1.6
Females						
GLU (mg/dL)	118.6 ± 21.38	120.2 ± 26.76	130.2 ± 21.58	121.4 ± 10.78	138.7 ± 26.58	157.8 ± 28.10
BUN (mg/dL)	18.2 ± 3.44	17.2 ± 3.22	18.1 ± 2.94	18.1 ± 2.23	16.5 ± 0.84	16.7 ± 2.92
CREA (mg/dL)	0.69 ± 0.053	0.69 ± 0.073	0.72 ± 0.122	0.69 ± 0.031	0.66 ± 0.042	0.65 ± 0.035
TP (g/dL)	7.55 ± 0.282	7.32 ± 0.475	7.57 ± 0.489	7.84 ± 0.434	7.40 ± 0.368	7.81 ± 0.532
ALB (g/dL)	4.80 ± 0.175	4.71 ± 0.261	4.85 ± 0.218	5.00 ± 0.253	4.76 ± 0.200	5.07 ± 0.327
A/G (ratio)	1.75 ± 0.061	1.82 ± 0.097	1.80 ± 0.124	1.77 ± 0.140	1.82 ± 0.154	1.85 ± 0.127
TCHO (mg/dL)	76.7 ± 22.37	72.5 ± 12.48	70.6 ± 15.47	74.4 ± 10.77	86.8 ± 22.62	72.8 ± 21.02
TG (mg/dL)	43.0 ± 14.97	42.5 ± 12.79	39.0 ± 11.91	43.2 ± 9.17	53.0 ± 18.36	63.8 ± 22.29
PL (md/dL)	139 ± 33.4	132 ± 20.3	135 ± 25.5	145 ± 17.4	152 ± 24.8	140 ± 30.3
AST (IU/L)	133.4 ± 24.86	115 ± 18.37	114.2 ± 19.30	163 ± 74.02	144.8 ± 50.18	113.3 ± 20.55
ALT (IU/L)	44.3 ± 19.04	37 ± 10.41	34.9 ± 11.75	63.7 ± 53.63	43.6 ± 21.30	38.7 ± 18.84
TBIL (mg/dL)	0.130 ± 0.0286	0.134 ± 0.0159	0.148 ± 0.0315	0.141 ± 0.0330	0.147 ± 0.0229	0.140 ± 0.0302
ALP (IU/L)	154.4 ± 50.00	158.3 ± 23.39	123.4 ± 35.28	139.0 ± 21.99	113.8 ± 15.00	95.7 ± 4.77*
CK (IU/L)	688 ± 135.0	485 ± 202.8*	468 ± 97.3*	587 ± 203.0	745 ± 211.5	580 ± 172.8
Ca (mg/dL)	11.56 ± 0.307	11.63 ± 0.433	11.78 ± 0.374	11.76 ± 0.343	12.31 ± 0.351	12.50 ± 0.403
IP (mg/dL)	7.17 ± 1.202	7.47 ± 1.152	7.21 ± 1.127	7.18 ± 0.817	8.14 ± 0.385	7.35 ± 0.496*
Na (mmol/L)	145 ± 1.2	145 ± 1.4	146 ± 1.6	145 ± 2.3	147 ± 1.9	146 ± 1.1
K (mmol/L)	7.14 ± 0.765	7.60 ± 0.895	7.36 ± 0.908	7.40 ± 1.167	7.50 ± 1.324	6.49 ± 0.501
Cl (mmol/L)	103 ± 1.9	103 ± 1.0	103 ± 1.3	103 ± 1.9	105 ± 1.9	101 ± 0.5**

Values are means ± S.D.

(R): Recovery period.

*: Significant difference compared with control ($p < 0.05$).**: Significant difference compared with control ($p < 0.01$).

3.2. Animal observations

Neither deaths nor treatment-related clinical signs were observed in any of the groups (Tables 1 and 2). Mean body weight gain and final mean body weight of male and female rats exposed to increasing doses of Korean red ginseng extract for 13 weeks were not significantly different from those of the sterile distilled water controls (Fig. 1). No remarkable difference was observed in the overall average of food consumption between the control and treated groups in either sex (Table 3). From ophthalmological examination, no treatment-related abnormalities were observed in any group of either sex (data not shown).

3.3. Hematology and serum biochemistry

The results of hematological analyses are shown in Table 4. Monocytes for main group males on 1000 mg/kg/day were significantly increased. Furthermore, statistically significant increases in PT were observed in the main group males on 1000 mg/kg/day and in the recovery group males on 2000 mg/kg/day. In females, there were no significant differences in any hematological parameters between the control and the treated groups. In serum biochemical examinations, statistically significant decreases for IP in the main group males on 500 and 1000 mg/kg/day, and CK in the main group females on 500 and 1000 mg/kg/day were observed (Table 5). In addition, decreases of ALP, IP, and Cl were also noted in recovery group females on 2000 mg/kg/day.

Table 6
Terminal absolute and relative organ weights observed at the conclusion of 13-week and 4-week recovery studies.

Dose (mg/kg/day)			Con			500			1000			2000			Con (R)			2000 (R)		
Sex and Organ	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)		
Male																				
Brain	2.136 ± 0.10-42	0.411 ± 0.02-89	2.077 ± 0.06-65	0.414 ± 0.03-45	2.144 ± 0.13-31	0.421 ± 0.04-25	2.214 ± 0.11-87	0.400 ± 0.02-03	2.152 ± 0.58-88	0.39 ± 0.016-1	2.145 ± 0.06-27	0.383 ± 0.-0417								
Pituitary gland	0.011 ± 0.00-16	0.002 ± 0.00-03	0.012 ± 0.00-11	0.002 ± 0.00-02	0.011 ± 0.00-18	0.002 ± 0.00-03	0.012 ± 0.00-19	0.002 ± 0.00-04	0.013 ± 0.00-16	0.002 ± 0.00-03	0.012 ± 0.00-08	0.002 ± 0.-0003								
Liver	14.016 ± 1.6-318	2.686 ± 0.18-56	13.048 ± 1.7-631	2.574 ± 0.17-13	14.480 ± 2.3-656	2.809 ± 0.18-56	14.914 ± 1.3-574	2.804 ± 0.13-98	14.133 ± 1.5-092	2.559 ± 0.22-09	15.082 ± 1.8-634	2.665 ± 0.-1268								
Spleen	0.775 ± 0.07-57	0.149 ± 0.01-46	0.769 ± 0.06-52	0.152 ± 0.01-11	0.773 ± 0.13-98	0.150 ± 0.01-91	0.836 ± 0.10-06	0.157 ± 0.01-73	0.797 ± 0.11-08	0.144 ± 0.01-49	0.823 ± 0.14-13	0.145 ± 0.-0107								
Heart	1.530 ± 0.09-55	0.294 ± 0.01-92	1.529 ± 0.21-93	0.302 ± 0.02-55	1.502 ± 0.19-41	0.293 ± 0.02-35	1.516 ± 0.13-20	0.285 ± 0.01-65	1.566 ± 0.13-80	0.285 ± 0.03-47	1.586 ± 0.21-63	0.281 ± 0.-0107								
Thymus	0.383 ± 0.07-56	0.073 ± 0.01-11	0.345 ± 0.06-87	0.068 ± 0.01-06	0.393 ± 0.10-21	0.077 ± 0.01-83	0.401 ± 0.11-55	0.075 ± 0.01-89	0.379 ± 0.05-90	0.069 ± 0.01-11	0.361 ± 0.05-60	0.064 ± 0.-0070								
Salivary glands	0.706 ± 0.06-73	0.136 ± 0.01-42	0.777 ± 0.07-60	0.154 ± 0.01-56	0.707 ± 0.10-46	0.139 ± 0.02-23	0.766 ± 0.06-70	0.145 ± 0.01-64	0.718 ± 0.03-60	0.130 ± 0.00-49	0.726 ± 0.03-30	0.129 ± 0.-0123								
Seminal vesicle	2.026 ± 0.12-91	0.390 ± 0.03-42	2.018 ± 0.31-70	0.402 ± 0.07-10	2.082 ± 0.40-96	0.409 ± 0.08-69	2.216 ± 0.34-77	0.402 ± 0.07-40	2.031 ± 0.24-17	0.370 ± 0.05-33	1.806 ± 0.14-20	0.324 ± 0.-0574								
Prostate	0.645 ± 0.13-68	0.125 ± 0.28-4	0.615 ± 0.09-30	0.123 ± 0.02-40	0.645 ± 0.18-20	0.126 ± 0.03-27	0.703 ± 0.17-45	0.133 ± 0.03-57	0.687 ± 0.20-97	0.124 ± 0.03-49	0.675 ± 0.07-99	0.120 ± 0.-0124								
Kidneys	3.548 ± 0.22-58	0.682 ± 0.03-42	3.601 ± 0.41-48	0.712 ± 0.04-29	3.553 ± 0.48-98	0.692 ± 0.05-58	3.828 ± 0.22-22	0.723 ± 0.05-85	3.717 ± 0.18-63	0.674 ± 0.01-37	3.775 ± 0.29-93	0.670 ± 0.-0465								
Adrenal glands	0.058 ± 0.00-91	0.011 ± 0.00-20	0.058 ± 0.00-42	0.011 ± 0.00-10	0.058 ± 0.00-92	0.011 ± 0.00-14	0.061 ± 0.00-65	0.012 ± 0.00-11	0.057 ± 0.00-37	0.010 ± 0.00-09	0.060 ± 0.00-58	0.011 ± 0.-0011								
Testes	3.361 ± 0.15-13	0.648 ± 0.04-68	3.550 ± 0.30-92	0.705 ± 0.06-58	3.306 ± 0.30-33	0.652 ± 0.09-56	3.534 ± 0.25-72	0.668 ± 0.07-20	3.334 ± 0.26-95	0.605 ± 0.05-04	3.536 ± 0.43-12	0.632 ± 0.-1145								
Epididymides	1.534 ± 0.08-64	0.296 ± 0.02-42	1.614 ± 0.16-98	0.320 ± 0.02-88	1.535 ± 0.12-19	0.302 ± 0.03-80	1.506 ± 0.12-83	0.284 ± 0.02-64	1.564 ± 0.14-82	0.283 ± 0.02-16	1.599 ± 0.15-12	0.286 ± 0.-0451								
Lung	1.642 ± 0.09-43	0.316 ± 0.01-59	1.615 ± 0.12-16	0.320 ± 0.01-80	1.598 ± 0.17-40	0.313 ± 0.02-74	1.712 ± 0.17-41	0.323 ± 0.03-58	1.767 ± 0.13-70	0.321 ± 0.03-72	1.798 ± 0.14-94	0.319 ± 0.-0139								
Thyroid/ Parathy.	0.027 ± 0.00-35	0.005 ± 0.00-07	0.023 ± 0.00-34	0.005 ± 0.00-05	0.025 ± 0.00-45	0.005 ± 0.00-08	0.024 ± 0.00-49	0.005 ± 0.00-08	0.028 ± 0.00-48	0.005 ± 0.00-08	0.020 ± 0.00-28*	0.004 ± 0.-0007*								
Female																				
Brain	1.944 ± 0.10-05	0.716 ± 0.06-79	1.931 ± 0.10-16	0.699 ± 0.07-95	1.935 ± 0.09-82	0.700 ± 0.04-79	1.937 ± 0.05-96	0.683 ± 0.04-33	2.007 ± 0.08-42	0.640 ± 0.05-86	1.939 ± 0.07-94	0.640 ± 0.-0670								
Pituitary gland	0.016 ± 0.00-29	0.006 ± 0.00-09	0.015 ± 0.00-30	0.006 ± 0.00-10	0.017 ± 0.00-29	0.006 ± 0.00-09	0.015 ± 0.00-19	0.005 ± 0.00-08	0.017 ± 0.00-17	0.005 ± 0.00-07	0.015 ± 0.00-24	0.005 ± 0.-0011								
Liver	7.320 ± 1.12-95	2.671 ± 0.26-24	7.363 ± 0.74-44	2.643 ± 0.13-70	7.681 ± 1.10-58	2.763 ± 0.28-88	8.046 ± 0.64-66	2.834 ± 0.21-84	8.127 ± 0.87-49	2.576 ± 0.15-56	8.366 ± 0.60-31	2.756 ± 0.-2539								
Spleen	0.522 ± 0.10-70	0.191 ± 0.03-57	0.527 ± 0.08-10	0.190 ± 0.03-28	0.496 ± 0.09-91	0.178 ± 0.02-75	0.535 ± 0.14-82	0.188 ± 0.05-19	0.561 ± 0.09-08	0.178 ± 0.02-31	0.514 ± 0.06-12	0.168 ± 0.-0105								
Heart	0.907 ± 0.10-29	0.333 ± 0.03-17	0.897 ± 0.10-40	0.321 ± 0.01-45	0.930 ± 0.11-40	0.335 ± 0.03-08	0.939 ± 0.05-44	0.331 ± 0.01-82	1.070 ± 0.12-96	0.339 ± 0.02-72	1.022 ± 0.09-14	0.336 ± 0.-0298								
Thymus	0.302 ± 0.05-47	0.111 ± 0.02-34	0.296 ± 0.04-66	0.107 ± 0.01-70	0.285 ± 0.04-43	0.103 ± 0.01-45	0.291 ± 0.04-53	0.102 ± 0.01-52	0.338 ± 0.07-19	0.106 ± 0.01-10	0.255 ± 0.06-90	0.082 ± 0.-0131*								
Salivary glands	0.462 ± 0.04-86	0.169 ± 0.01-04	0.450 ± 0.02-55	0.162 ± 0.01-42	0.442 ± 0.03-66	0.160 ± 0.01-55	0.464 ± 0.03-77	0.164 ± 0.01-81	0.488 ± 0.03-12	0.155 ± 0.01-36	0.454 ± 0.03-35	0.149 ± 0.-0116								
Kidneys	1.953 ± 0.19-65	0.715 ± 0.04-14	2.016 ± 0.24-30	0.723 ± 0.04-42	2.061 ± 0.24-68	0.743 ± 0.07-42	2.065 ± 0.16-89	0.727 ± 0.04-57	2.194 ± 0.25-89	0.694 ± 0.03-36	2.265 ± 0.27-76	0.744 ± 0.-0761								
Adrenal glands	0.074 ± 0.00-95	0.027 ± 0.00-40	0.070 ± 0.00-89	0.025 ± 0.00-28	0.076 ± 0.00-83	0.027 ± 0.00-29	0.076 ± 0.00-78	0.027 ± 0.00-39	0.080 ± 0.18-6	0.025 ± 0.00-37	0.079 ± 0.00-54	0.026 ± 0.-0032								

(continued on next page)

Table 6 (continued)

Sex and Organ	Con		500		1000		2000		Con (R)		2000 (R)	
	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)	Absolute (g)	Relative (%)
Ovaries	0.079 ± 0.01-17	0.029 ± 0.00-34	0.086 ± 0.01-02	0.031 ± 0.00-28	0.094 ± 0.02-13	0.034 ± 0.00-61	0.089 ± 0.01-70	0.031 ± 0.00-65	0.087 ± 0.01-44	0.028 ± 0.00-32	0.090 ± 0.01-22	0.030 ± 0.00-45
Lung	1.181 ± 0.08-88	0.434 ± 0.02-93	1.198 ± 0.09-20	0.431 ± 0.03-11	1.187 ± 0.09-59	0.428 ± 0.02-76	1.224 ± 0.14-44	0.430 ± 0.03-35	1.412 ± 0.20-41	0.449 ± 0.06-81	1.286 ± 0.15-06	0.422 ± 0.03-79
Thyroid/Parathy.	0.018 ± 0.00-40	0.006 ± 0.00-13	0.018 ± 0.00-46	0.006 ± 0.00-14	0.019 ± 0.00-33	0.007 ± 0.00-13	0.019 ± 0.00-37	0.007 ± 0.00-11	0.020 ± 0.00-42	0.006 ± 0.00-09	0.020 ± 0.00-86	0.007 ± 0.00-02
Uterus/Cervix	0.816 ± 0.23-47	0.302 ± 0.09-27	0.731 ± 0.19-05	0.266 ± 0.07-74	0.704 ± 0.20-50	0.255 ± 0.07-57	0.810 ± 0.27-82	0.283 ± 0.09-32	0.725 ± 0.08-69	0.232 ± 0.03-85	0.606 ± 0.05-32*	0.200 ± 0.02-31

Values are means ± S.D.

(R): Recovery period.

*: Significant difference compared with control ($p < 0.05$).

3.4. Urinalysis

From urinalysis, no treatment-related changes were observed in any group of either sex (Data not shown).

3.5. Necropsy and organ weight

In gross findings, abnormal shape and focus/foci of the liver were observed in 1 main group male on 1000 mg/kg/day, and a markedly small thyroid/parathyroid was observed in 1 recovery group male on 2000 mg/kg/day. In females, a slightly small thyroid/parathyroid was observed in 2 cases in the main group on 500 mg/kg/day, and a minimally enlarged spleen was observed in 1 case in the main group on 2000 mg/kg/day. Table 6 summarizes the terminal absolute and relative organ weights observed at the conclusion of the 13-week and recovery studies. In recovery group males, statistically significant decreases in absolute and relative thyroid/parathyroid weights were observed in the 2000 mg/kg/day group. In addition, statistically significant decreases in absolute uterus/cervix weight and in relative thymus weight were noted in recovery group females in the 2000 mg/kg/day group.

3.6. Histopathology

From histopathological examination, no treatment-related abnormal findings were observed in any group of either sex (Fig. 2, Table 7).

4. Discussion

Although safety issues remain to be clarified regarding specific preparations, ginseng is presumed to be a nontoxic herbal medicine. Because, the ginseng preparations used for safety tests are not standardized, there are limitations in comparing different safety tests that vary dose levels of each preparation.

To investigate the repeated oral dose toxicity of Korean red ginseng extract, the test article was administered orally once a day to male and female SD rats at dose levels of 0, 500, 1000, or 2000 mg/kg/day for 13 consecutive weeks. Study parameters included mortality, clinical observations, body weight changes, food consumption, ophthalmologic examinations, hematology, serum biochemistry, necropsy, organ weight, and histopathological examinations.

Loss of fur, scratch wounds, scabs, hard skin, eye discharge, and loss of teeth were among the clinical symptoms observed in some animals. However, these were considered to be unrelated to treatment, since they might be generally observed during maintenance of animals, and were not dose-dependent.

Hematological and serum biochemistry profiles showed statistically significant increases in monocytes and prothrombin times, and decreases in IP, CK, ALP, and Cl in the main group of males and females and the recovery-group males. However, these changes were considered to be unrelated to treatment because they were not dose-dependent, and were unaccompanied by correlative findings. Anatomical abnormalities in the main group of males included abnormal shapes and focus/foci of livers. However, they were not considered to be of toxicological significance because of the lack of histopathological disorders, and the absence of any increases in ALT, ALP, and AST. The absolute size of each thyroid/parathyroid in the recovery-group males and main-group females was observed to be small, and the absolute and relative thyroid/parathyroid weights decreased in the recovery-group males. However, no obvious abnormalities were observed in thyroid/

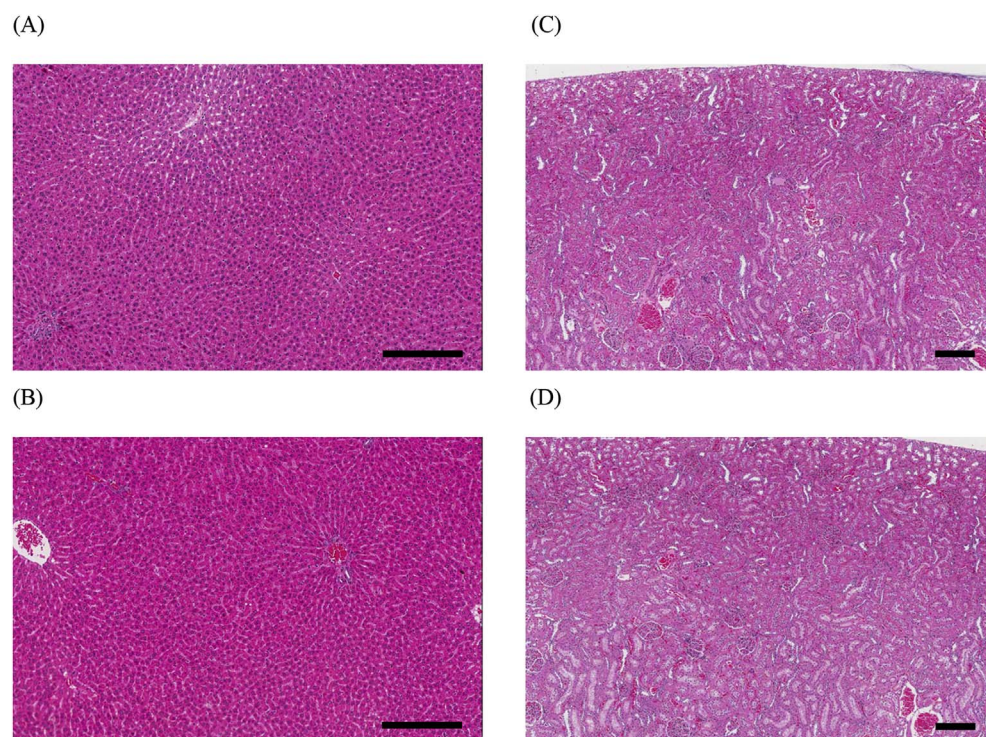


Fig. 2. Histopathology of the liver (A and B) and kidneys (C and D) of male rats administrated Korean red ginseng extract for 13 weeks. Panels A and C: 0 mg/kg/day group; Panels B and D: 2000 mg/kg/day. HE staining. Scale bar: 250µm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

parathyroid histopathology, indicating that the changes in the thyroid/parathyroid were not toxicologically significant. An enlarged spleen was observed in the main-group females administered of ginseng preparation 2000 mg/kg/day, but it was considered not to be of toxicological significance, since the change was within the normal physiologic range and the histopathology of the spleen was normal. From organ weight measurements, decreases in absolute uterus/cervix and in relative thymus of the recovery-group females were considered to be unrelated to treatment, because the same changes were not observed in the main group. No treatment-related abnormal findings were observed in any of the groups following histopathological examinations. It was considered that the severity of other changes found was minimal and also occurred spontaneously in similarly aged SD rats.

Ginseng is considered to be comparatively safe, there are reported cases of ginseng preparation safety tests in animals. Beagle dogs fed diets containing ginseng preparations at dose levels of 15 mg/kg (0.6 mg ginsenoside/kg) for 90 days showed no effect of toxicity (Hess et al., 1983). Additionally, rats taking a ginseng preparation at daily dose levels of 105–210 mg/kg for 25 weeks presented no symptoms attributable to toxicity (Popov and Goldwag, 1973). In the present study, we administrated Korean red ginseng preparation (2000 mg/kg/day) for 13 weeks, but there were no toxic side effects in male or female rats. The present study corresponds with a preliminary, short-term, subacute oral toxicity study (four weeks) of Korean red ginseng extract in SD rats, which showed no toxicity (Park et al., 2013). These animal studies consistently support the safety of ginseng preparations; however, the ginseng preparations

used for the tests were not standardized, indicating that the composition and/or ratio of the effective components such as ginsenosides could differ during preparation or manufacturing processes. In this study, experiments were performed under controlled conditions, satisfying the GLP with a standardized ginseng preparation i.e., clearly defined and controlled constitution of the ginsenosides by a precisely established manufacturing process. In order to understand ginseng preparation toxicity more thoroughly, further studies are required, using a standardized ginseng preparation and performed examinations under controlled conditions.

The recovery period i.e., a non-dosing period that follows the main dosing phase of a study, was designed to understand whether toxicities observed at the end of the dosing phase are partially or completely reversible. In addition, it was also used to determine whether the test sample has the potential to produce delayed toxicity after the dosing administration is done. However, no toxicity was observed during the dosing period and no delayed toxicity was observed during the recovery period.

In conclusion, after 13 consecutive weeks of daily oral dosing with Korean red ginseng extract in SD rats at dose levels of 0, 500, 1000, or 2000 mg/kg/day, there were no treatment-related changes in both sexes of the main and recovery groups. Therefore, it was concluded that the no-observed adverse effect level (NOAEL) of the test sample was established to be 2000 mg/kg/day for both sexes in this study.

Table 7

Histopathological findings of male and female SD rats orally administered with Korean red ginseng extract for 13-week and 4-week recovery studies.

Sex	Male				Female			
Dose (mg/kg/day)	Con	2000	Con (R)	2000 (R)	Con	2000	Con (R)	2000 (R)
Number examined	10	10	5	5	10	10	5	5
Tissue with diagnoses								
Kidney								
Tubular degeneration/regeneration	4	3	3	0	1	0	0	0
Infiltration, mononuclear cell	5	3	0	2	2	0	0	1
Casts	0	2	0	0	1	0	0	0
Mineralization	0	0	0	0	2	2	1	1
Pelvic dilatation	0	0	0	1	0	0	0	0
Liver								
Infiltration, mononuclear cell	5	5	4	2	8	3	3	2
Vacuolated area/focus	3	0	2	2	0	2	3	1
Focal necrosis	1	1	1	0	0	0	0	2
Bile duct proliferation	3	0	1	1	1	0	0	0
Capsular fibrosis	1	0	0	0	0	0	0	0
Squamous cyst	0	1	0	0	0	0	0	0
Infiltration, granulomatous	0	1	0	0	0	0	0	0
Hepatocyte vacuolation	0	2	0	0	0	0	0	0
Heart								
Cardiomyopathy	0	1	1	1	1	0	0	0
Infiltration, mononuclear cell	0	0	0	1	1	0	0	0
Pancreas								
Acinar cell atrophy	3	1	0	0	0	0	0	0
Infiltration, mononuclear cell	2	2	2	2	2	0	0	2
Islet fibrosis	0	0	0	1	0	0	0	0
Lung								
Aggregation, foamy macrophage	2	2	1	0	1	1	0	0
Eosinophilic crystals	0	1	0	0	0	0	0	0
Osseous metaplasia	0	0	0	0	1	1	0	0
Adrenal glands								
Cortical vacuolation	2	3	3	2	0	0	0	0
Cortical hypertrophy	0	1	0	0	0	0	0	0
Harderian glands								
Infiltration, mononuclear cell	0	1	0	0	1	1	1	0
Stomach								
Squamous hyperplasia	0	1	0	0	0	0	0	0
Rectum								
Lymphoid hyperplasia	0	0	0	0	1	0	0	0
Skin								
Epidermal hyperplasia	0	1	0	0	0	0	0	0
Crust	0	1	0	0	0	0	0	0
Femur joint marrow								
Arteriopathy	0	1	0	0	0	0	0	0
Vagina								
Proestrus					6	4	1	0
Estrus					2	3	1	1
Metestrus					2	2	2	1
Diestrus					0	1	1	3

(R): Recovery period.

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Transparency document

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